

Determine the MMS guarantee of EFX pure Nash equilibria for two agents

ChatGPT (gpt-5.6-sol)*

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Abstract

We consider mechanisms without payments for allocating indivisible goods between two strategic agents with nonnegative additive valuations. A mechanism is called *qualifying* if every valuation instance admits a pure Nash equilibrium and every pure Nash equilibrium induces an allocation that is envy-free up to any positively valued good (EFX) with respect to the agents' true valuations.

Previous work showed that every equilibrium of every qualifying mechanism provides strictly more than two thirds of each agent's maximin share, but did not obtain a uniform improvement over two thirds. We determine the exact uniform guarantee: if

$$\alpha^* = \inf_{\substack{\Phi \text{ qualifying, } M, v, \\ b \in \text{PNE}_\Phi(v), i: \mu_i > 0}} \frac{v_i(\Phi(b)_i)}{\mu_i},$$

then

$$\alpha^* = \frac{5}{6}.$$

The lower bound follows from a menu argument: fixing the other agent's report produces a menu whose demanded bundles must be EFX for every nonnegative additive objective. We prove that every value-maximizing bundle in such a menu is necessarily a 5/6-approximation to the two-way maximin share. For tightness, we give a five-good mechanism for which every instance has a pure Nash equilibrium and every pure Nash equilibrium is EFX, together with an equilibrium attaining exactly 5/6 of one agent's maximin share.

1 The open problem

This paper addresses an open problem posed by Georgios Amanatidis, Georgios Birmas, Federico Fusco, Philip Lazos, Stefano Leonardi, Rebecca Reiffenhäuser in *Allocating Indivisible*

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260826_175300_814f3d33`, `sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

Goods to Strategic Agents: Pure Nash Equilibria and Fairness [1] (Mathematics of Operations Research, 2023), Page 22, Section 5 (Discussion). The website catalogue entry is problem 99 (w4389193896_p0) of the [Open Problems in Operations Research](#) collection.

Find the best universal constant $\alpha^ \in [0, 1]$ such that the following holds:*

For every mechanism $\mathcal{M}$ that, for every additive-valuation instance (v_1, v_2) , has at least one pure Nash equilibrium and every pure Nash equilibrium induces an EFX allocation with respect to (v_1, v_2) , every such equilibrium allocation $A = \mathcal{M}(b)$ satisfies

$$v_i(A_i) \geq \alpha^* \mu_i \quad \text{for both } i \in \{1, 2\}.$$

Equivalently, determine the tight worst-case MMS approximation guarantee (as a function-free constant) enjoyed by EFX allocations arising at PNE of mechanisms whose PNE always exist and are always EFX. In particular, decide whether one can always take $\alpha^ = 1$ (MMS always), or whether $\alpha^* < 1$, and if so, determine the optimal α^* (the paper proves existence of some $\alpha > 2/3$).*

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

For two agents, an arbitrary EFX allocation guarantees two thirds of each agent’s maximin share, and this factor is tight without strategic restrictions. The strategic setting, however, imposes additional structure. In particular, Amanatidis et al. [2] showed that if a mechanism has pure Nash equilibria for every two-agent instance and all of its equilibria are EFX with respect to the true valuations, then every such equilibrium gives each agent strictly more than two thirds of her maximin share. Their argument is pointwise: the amount by which the ratio exceeds $2/3$ may depend on the instance and the equilibrium.

We show that the strategic separation is uniform and sharp: the exact worst-case ratio over all qualifying mechanisms, instances, equilibria, and agents is $5/6$.

The proof has two parts. For the lower bound, fixing the other agent’s report turns the mechanism into a menu of attainable bundles. A zero-valuation construction shows that every bundle demanded from this menu, under every nonnegative additive objective, must be EFX. We then prove a sharp structural lemma for such menus.

For tightness, we construct a mechanism on five cyclically ordered goods. For a fixed report of agent 2, the attainable bundles of agent 1 form either the edge set of a five-cycle or a knapsack menu

$$\{B : w(B) \leq \min\{w(M)/2, h\}\},$$

where h is the maximum reported weight of a cycle edge. This menu is rich enough to guarantee equilibrium existence, yet structured enough to make every equilibrium EFX. A concrete equilibrium has maximin-share ratio exactly $5/6$.

3 Model and benchmark

Let M be a finite set of indivisible goods and let $N = \{1, 2\}$ be the set of agents. Agent i has a nonnegative additive valuation $v_i : 2^M \rightarrow \mathbb{R}_{\geq 0}$, given by

$$v_i(S) = \sum_{g \in S} v_i(g).$$

An allocation is a partition $A = (A_1, A_2)$ of M .

Definition 1 (EFX). An allocation $A = (A_1, A_2)$ is *envy-free up to any positively valued good*, or *EFX*, if for every pair $i, j \in \{1, 2\}$ and every $g \in A_j$ with $v_i(g) > 0$,

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}).$$

For a nonnegative additive valuation v and a set S , write

$$\mu_v(2, S) = \max_{X \subseteq S} \min\{v(X), v(S \setminus X)\}.$$

When $S = M$, we abbreviate this as μ_v ; for agent i , we write $\mu_i = \mu_{v_i}$.

A mechanism Φ receives reports $b_i \in \mathbb{R}_{\geq 0}^M$ and returns an allocation $\Phi(b_1, b_2) = (\Phi(b)_1, \Phi(b)_2)$. Utilities are evaluated using the true valuations.

A report profile $b = (b_1, b_2)$ is a pure Nash equilibrium (PNE) for the true valuation profile $v = (v_1, v_2)$ if

$$v_i(\Phi(b)_i) \geq v_i(\Phi(b'_i, b_{-i})_i)$$

for every agent i and every alternative report b'_i .

Definition 2 (Qualifying mechanism). A mechanism, or a mechanism family defined for every finite ground set, is *qualifying* if:

1. every two-agent additive instance admits at least one pure Nash equilibrium; and
2. every pure Nash equilibrium induces an EFX allocation with respect to the agents' true valuations.

Let $\mathfrak{Q}$ denote the class of qualifying mechanisms. We study the universal worst-case guarantee

$$\alpha^* = \inf_{\substack{\Phi \in \mathfrak{Q}, M, v, \\ b \in \text{PNE}_\Phi(v), i: \mu_i > 0}} \frac{v_i(\Phi(b)_i)}{\mu_i}. \quad (1)$$

Remark. The order of quantifiers in (1) is important: it asks for a guarantee shared by *every* qualifying mechanism. If one instead maximized over the choice of mechanism, the answer would be 1, since Mod-Cut&Choose has only MMS equilibria [2].

4 Demand-EFX menus

Fix a ground set M . A nonempty family $\mathcal{D} \subseteq 2^M$ will be viewed as a menu of attainable bundles.

Definition 3 (Demand-EFX menu). A menu $\mathcal{D} \subseteq 2^M$ is *demand-EFX* if, for every nonnegative additive valuation w , every bundle

$$D \in \arg \max_{B \in \mathcal{D}} w(B)$$

is EFX against its complement:

$$w(D) \geq w(M \setminus (D \cup \{g\})) \quad \text{for every } g \in M \setminus D \text{ with } w(g) > 0.$$

We first record the standard two-thirds estimate in the form needed below.

Lemma 1 (EFX implies two-thirds MMS). *Let v be nonnegative and additive. If $A \subseteq M$ is EFX against $M \setminus A$, then*

$$v(A) \geq \frac{2}{3} \mu_v.$$

Proof. Let $B = M \setminus A$, and let k be the number of goods in B with positive v -value.

If $k \leq 1$, then in every bipartition of M , the part that does not contain the sole positively valued good of B , if there is one, is contained in A . Hence $\mu_v \leq v(A)$.

Suppose $k \geq 2$. EFX gives

$$v(B) - v(g) \leq v(A)$$

for each positively valued $g \in B$. Summing these k inequalities yields

$$(k-1)v(B) \leq k v(A),$$

and hence $v(B) \leq 2v(A)$. Therefore

$$2\mu_v \leq v(M) = v(A) + v(B) \leq 3v(A),$$

which proves the claim. $\square$

The strategic content of the problem is captured by the following reduction.

Lemma 2 (Strategic-menu reduction). *Let Φ be qualifying. Fix an agent i and a report b_{-i} of the other agent, and define*

$$\mathcal{D}_i(b_{-i}) = \{B \subseteq M : B = \Phi(r_i, b_{-i})_i \text{ for some } r_i \in \mathbb{R}_{\geq 0}^M\}.$$

Then $\mathcal{D}_i(b_{-i})$ is demand-EFX.

Moreover, if b is a PNE for the true valuation profile v , then

$$\Phi(b)_i \in \arg \max_{B \in \mathcal{D}_i(b_{-i})} v_i(B).$$

Proof. Let w be a nonnegative additive valuation, and let

$$B \in \arg \max_{D \in \mathcal{D}_i(b_{-i})} w(D).$$

Choose a report r_i attaining B , and consider the true valuation profile in which agent i has valuation w and the other agent has the identically zero valuation.

Agent i is best-responding because B maximizes w over every bundle attainable against b_{-i} . The other agent is also best-responding, since every possible deviation gives her utility zero. Thus (r_i, b_{-i}) is a genuine PNE. Since Φ is qualifying, its allocation is EFX; in particular, B is EFX for w , proving that the menu is demand-EFX.

The second assertion is simply the best-response condition at the original equilibrium b . $\square$

We now prove the main structural lemma.

Lemma 3 (Sharp menu lemma). *Let $\mathcal{D} \subseteq 2^M$ be demand-EFX, let v be nonnegative and additive, and let*

$$A \in \arg \max_{D \in \mathcal{D}} v(D).$$

Then

$$v(A) \geq \frac{5}{6} \mu_v.$$

Proof. We first remove a harmless technicality concerning zero-valued goods. Let

$$P = \{g \in M : v(g) > 0\}$$

and project the menu to

$$\mathcal{D}|_P = \{D \cap P : D \in \mathcal{D}\}.$$

Any auxiliary valuation on P can be extended as zero on $M \setminus P$, so $\mathcal{D}|_P$ remains demand-EFX. Moreover,

$$\mu_v(2, P) = \mu_v(2, M) \quad \text{and} \quad v(A \cap P) = v(A).$$

We may therefore replace M , $\mathcal{D}$, and A by their projections and assume that every good has positive v -value.

By Lemma 1, $v(A) \geq (2/3)\mu_v$; in particular, if $\mu_v > 0$, then $v(A) > 0$. Normalize

$$v(A) = 1.$$

Suppose for a contradiction that

$$\mu_v > \frac{6}{5}.$$

Put

$$C = M \setminus A, \quad y = v(C).$$

Since $2\mu_v \leq v(M) = 1 + y$,

$$y > \frac{7}{5}. \tag{2}$$

Since A is demanded under v , it is EFX. Thus for every $c \in C$,

$$1 \geq y - v(c),$$

and hence

$$v(c) \geq y - 1 > \frac{2}{5}. \tag{3}$$

There cannot be four or more goods in C : if $|C| = k \geq 4$, then summing (3) in the equivalent form $v(c) \geq y - 1$ gives

$$y \geq k(y - 1),$$

so $y \leq k/(k - 1) \leq 4/3$, contradicting (2). There also cannot be only one good in C , since then every bipartition would have a part contained in A , giving $\mu_v \leq 1$. Consequently,

$$|C| \in \{2, 3\}. \tag{4}$$

Transition bundles. For every $c \in C$, choose a bundle $D_c \in \mathcal{D}$ that maximizes $v(D)$ among menu bundles containing c , and write

$$\rho_c = v(D_c).$$

Such a bundle exists: otherwise, increasing only the auxiliary value of c would leave A demanded; since $|C| \geq 2$, we may choose $d \in C \setminus \{c\}$, and for a sufficiently large value on c , the EFX condition for A after the removal of d would fail.

Because A maximizes v over the menu,

$$\rho_c \leq 1.$$

Consider the auxiliary valuation

$$w_c = v + (1 - \rho_c)\mathbf{1}_c.$$

Every menu bundle not containing c has w_c -value at most 1. Every menu bundle containing c has v -value at most ρ_c , and therefore also has w_c -value at most 1. Both A and D_c have w_c -value exactly 1, so both are demanded.

Let

$$S_c = D_c \cap A.$$

EFX of D_c under w_c , after removing any $a \in A \setminus S_c$ from its complement, gives

$$1 \geq w_c(M \setminus (D_c \cup \{a\})) = 1 + y - \rho_c - v(a).$$

Thus

$$v(a) \geq y - \rho_c \geq y - 1 > \frac{2}{5} \quad (a \in A \setminus S_c). \quad (5)$$

EFX of A under w_c , after removing $d \in C \setminus \{c\}$, gives

$$1 \geq y - v(d) + 1 - \rho_c,$$

or

$$\rho_c \geq y - v(d) \quad (d \in C \setminus \{c\}). \quad (6)$$

We now treat the two possibilities in (4).

Case 1: two outside goods. Write

$$C = \{p, q\}, \quad p \geq q,$$

identifying goods with their values; thus $y = p + q > 7/5$.

Consider D_p . It cannot contain q , since otherwise $v(D_p) \geq p + q > 1$. Hence

$$D_p = \{p\} \cup S$$

for some $S \subseteq A$. Put

$$s = v(S), \quad R = A \setminus S.$$

Since $p + s \leq 1$,

$$s \leq 1 - p. \quad (7)$$

By (5), every $a \in R$ satisfies

$$v(a) \geq q - s > \frac{2}{5}.$$

Thus R has one or two goods.

Suppose first that $R = \{a\}$. In any bipartition whose two sides both have value greater than $6/5$, the goods p and q must lie on opposite sides, since $v(A) = 1$. One of those sides does not contain a and is therefore contained in $\{p\} \cup S$ or $\{q\} \cup S$; its value is at most

$$p + s \leq 1,$$

a contradiction.

Now suppose $R = \{a, b\}$. Then

$$1 - s = v(a) + v(b) \geq 2(q - s),$$

so

$$s \geq 2q - 1.$$

Since $p + q > 7/5$, we have $q > 7/5 - p$, and therefore

$$s > \frac{9}{5} - 2p.$$

Together with $s \leq 1 - p$, this implies

$$p > \frac{4}{5}.$$

In a bipartition with both sides above $6/5$, the goods p, q must be separated, and the two goods of R must also be separated; otherwise one side would be contained in one outside good together with S , of value at most $p + s \leq 1$.

Consider the side containing q . It contains at most one good of R and all or part of S . Since the other good of R has value at least $q - s$, the good of R placed with q has value at most

$$1 - s - (q - s) = 1 - q.$$

Consequently that side has value at most

$$q + (1 - q) + s = 1 + s \leq 2 - p < \frac{6}{5},$$

again a contradiction.

Thus $\mu_v \leq 6/5$ whenever $|C| = 2$.

Case 2: three outside goods. Write

$$C = \{p, q, r\}, \quad p \geq q \geq r.$$

EFX of A gives

$$p + q \leq 1, \quad p + r \leq 1, \quad q + r \leq 1. \tag{8}$$

Since $p + q + r > 7/5$, it follows that

$$q + r > \frac{4}{5}, \quad q - r < \frac{1}{10}, \quad p - r < \frac{1}{5}. \tag{9}$$

Indeed, $q \leq 1/2$, while $r > 2/5$, and $p - r \leq 1 - q - r < 1/5$.

The transition bounds (6) become

$$\rho_p \geq p + q, \quad \rho_q \geq p + q, \quad \rho_r \geq p + r. \quad (10)$$

We next determine the form of every transition bundle.

Claim 1. For each $c \in \{p, q, r\}$, the transition bundle D_c contains no other good of C .

Suppose instead that

$$D_c \cap C = T, \quad |T| = 2.$$

Let d be the remaining good of C , and write

$$S = D_c \cap A, \quad R = A \setminus S, \quad s = v(S).$$

Since $v(D_c) \leq 1$,

$$s \leq 1 - v(T).$$

Moreover, (5) gives

$$v(a) \geq d - s > \frac{2}{5} \quad (a \in R).$$

Hence $|R| \in \{1, 2\}$.

Suppose $|R| = 1$. In a hypothetical partition with both sides above $6/5$, the side containing the sole good of R must also contain a good of C ; otherwise it would be contained in A . Therefore the other side contains at most two goods of C , together with some subset of S , so its value is at most $p + q + s$. Depending on T , this is bounded as follows:

T	$p + q + s$
$\{p, q\}$	$\leq 1,$
$\{p, r\}$	$\leq 1 + q - r < \frac{11}{10},$
$\{q, r\}$	$\leq 1 + p - r < \frac{6}{5}.$

Thus no such partition exists.

Suppose $|R| = 2$. In a partition with both sides above $6/5$, the two goods of R must be separated. Indeed, if one side contains no good of R , then either it contains at most two goods of C , in which case the preceding table bounds its value below $6/5$, or it contains all three goods of C , in which case the other side is contained in A and has value at most 1.

The goods of C must also be split between the two sides. Consequently, one side contains one good of C , one good of R , and some subset of S . Since each good in R is worth at least $d - s$, the larger good of R is worth at most

$$1 - s - (d - s) = 1 - d.$$

The singleton- C side therefore has value at most

$$p + (1 - d) + s.$$

For all three possibilities for T , using $s \leq 1 - v(T)$, this is at most

$$2 - q - r < \frac{6}{5}.$$

This is again impossible, proving Claim 1.

By Claim 1,

$$D_c = \{c\} \cup S_c.$$

Put $s_c = v(S_c)$ and $R_c = A \setminus S_c$. Every good in R_c has value greater than $2/5$, so $|R_c| \leq 2$. If $|R_c| = 2$, then

$$1 - s_c \geq 2(y - c - s_c),$$

and hence

$$1 + s_c \geq 2(y - c).$$

Since $s_c \leq 1 - c$, this would imply

$$2y - c \leq 2.$$

But

$$2y - c = y + (y - c) > \frac{7}{5} + \frac{4}{5} > 2,$$

a contradiction. Thus R_c consists of a single good; denote it by a_c . We have

$$D_c = \{c\} \cup (A \setminus \{a_c\}).$$

Using (10), we obtain

$$p \leq a_p \leq 1 - q, \quad q \leq a_q \leq 1 - p, \quad r \leq a_r \leq 1 - p. \quad (11)$$

Let

$$a = a_p, \quad S = A \setminus \{a\}.$$

We now strengthen the first pair of bounds in (11).

Claim 2. The distinguished good a satisfies

$$p \leq a \leq 1 - p.$$

The lower bound is already contained in (11). Suppose, toward a contradiction, that

$$a > 1 - p.$$

Then $a \neq a_q, a_r$, so $a_q, a_r \in S$. If these were distinct goods, their combined value would exceed

$$q + r > \frac{4}{5},$$

whereas

$$v(S) = 1 - a < p < \frac{3}{5}.$$

Thus they must be the same good; write

$$b = a_q = a_r.$$

By (11),

$$q \leq b \leq 1 - p.$$

Let

$$T = S \setminus \{b\}, \quad t = v(T) = 1 - a - b.$$

Then

$$t < p - b \leq p - q < \frac{1}{5}. \quad (12)$$

Consider the five goods

$$K = \{a, b, p, q, r\}.$$

All five have value greater than $2/5$ and less than $3/5$. Furthermore, every pair from K , except possibly $\{a, p\}$, has value at most 1. Indeed,

$$a + q \leq 1, \quad b + p \leq 1, \quad a + b \leq 1,$$

and all remaining inequalities follow from these and (8).

In any bipartition with both sides above $6/5$, the five goods of K must therefore be split 2-against-3. The side containing two of them receives total residual value at most $t < 1/5$, so its pair from K would have to be worth more than 1. The pair must consequently be $\{a, p\}$. But even if that side receives all of T , its value is at most

$$a + p + t = 1 + p - b \leq 1 + p - q < \frac{6}{5},$$

contradicting the assumed partition. This proves Claim 2.

Since $p \leq a \leq 1 - p$, we have $p \leq 1/2$. Therefore

$$q + r = y - p > \frac{7}{5} - \frac{1}{2} = \frac{9}{10}. \quad (13)$$

Define

$$H = \{a, p, q, r\}.$$

The largest pair in H is $\{a, p\}$, and Claim 2 gives

$$v(T) \leq 1 \quad \text{for every } T \in \binom{H}{2}. \quad (14)$$

Let $(X, M \setminus X)$ be a bipartition with both sides above $6/5$, which exists because $\mu_v > 6/5$. A side containing at most one good of H has value at most

$$a + v(S) = a + (1 - a) = 1.$$

Thus each side contains exactly two goods of H . By (14), the S -part of each side has value greater than $1/5$. Those two S -parts partition S , so

$$\mu_v(2, S) > \frac{1}{5}. \quad (15)$$

We finish by extracting a smaller demand-EFX menu on S . Every good in H has value greater than $2/5$, while every menu bundle has v -value at most $v(A) = 1$. Hence no menu bundle contains three goods of H .

On the other hand, some menu bundle must contain two goods of H . Otherwise, give every good of H auxiliary value $L > 0$ and every good of S value zero. A demanded menu bundle would

contain at most one good of H , while its complement would contain at least three; removing one positively valued good from the complement would leave at least two goods of H , violating EFX.

Define the fibre family

$$\mathcal{F} = \left\{ R \subseteq S : T \cup R \in \mathcal{D} \text{ for some } T \in \binom{H}{2} \right\}.$$

This family is nonempty.

We claim that $\mathcal{F}$ is demand-EFX on the ground set S . Let u be any nonnegative additive valuation on S , and choose $L > u(S)$. Extend u to M by assigning value L to every good in H . Every demanded menu bundle then contains exactly two goods of H , and its S -part is precisely a u -maximizer in $\mathcal{F}$.

Let R be such an S -part. For every $g \in S \setminus R$ with $u(g) > 0$, global EFX gives

$$2L + u(R) \geq 2L + u(S \setminus (R \cup \{g\})).$$

After cancelling $2L$, this is exactly EFX inside S . Thus $\mathcal{F}$ is demand-EFX.

Take

$$R \in \arg \max_{X \in \mathcal{F}} v(X).$$

By Lemma 1 and (15),

$$v(R) \geq \frac{2}{3} \mu_v(2, S) > \frac{2}{15}. \quad (16)$$

By the definition of $\mathcal{F}$, there is a pair $T \in \binom{H}{2}$ such that $T \cup R \in \mathcal{D}$. Since $a \geq p \geq q \geq r$, every pair in H has value at least $q + r$. Because every menu bundle has value at most 1,

$$v(R) \leq 1 - v(T) \leq 1 - (q + r) < \frac{1}{10},$$

where the last inequality is (13). This contradicts (16).

Both cases in (4) lead to a contradiction. Hence $\mu_v \leq 6/5$ under the normalization $v(A) = 1$, or equivalently

$$v(A) \geq \frac{5}{6} \mu_v. \quad \square$$

5 The universal lower bound

Theorem 1. *Let Φ be any qualifying mechanism, let v be any two-agent additive instance, and let $b \in \text{PNE}_\Phi(v)$. Then, for both agents,*

$$v_i(\Phi(b)_i) \geq \frac{5}{6} \mu_i.$$

Consequently,

$$\alpha^* \geq \frac{5}{6}.$$

Proof. Fix an agent i and the equilibrium report b_{-i} of the other agent. By Lemma 2, the attainable menu $\mathcal{D}_i(b_{-i})$ is demand-EFX, and the equilibrium bundle $\Phi(b)_i$ maximizes v_i over this menu. The conclusion follows immediately from Lemma 3. $\square$

6 A tight five-good mechanism

We now construct a qualifying mechanism with an equilibrium attaining the bound.

Let

$$M = \{0, 1, 2, 3, 4\},$$

with indices interpreted modulo 5, and let

$$\mathcal{E} = \{\{0, 1\}, \{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 0\}\}$$

be the edge set of the cycle C_5 .

A *row* is a pair

$$(C, H) \in 2^M \times \mathcal{E}.$$

Fix any efficiently computable surjection

$$\sigma : \mathbb{R}_{\geq 0}^M \longrightarrow 2^M \times \mathcal{E}.$$

For example, enumerate the 160 rows as $R_0, \dots, R_{159}$, and set

$$\sigma(b_1) = R_{\lfloor b_1(0) \rfloor \bmod 160}.$$

Thus agent 1 can select any desired row through her report.

Define the mechanism Ψ as follows. Given reports (b_1, b_2) , let

$$(C, H) = \sigma(b_1).$$

1. If $b_2(M) = 0$, return

$$(A_1, A_2) = (H, M \setminus H).$$

2. If $b_2(M) > 0$, form the three labeled candidates for agent 2:

$$T_1 = C, \quad T_2 = M \setminus C, \quad T_3 = M \setminus H.$$

Give agent 2 a candidate maximizing b_2 , with fixed priority $T_1 \succ T_2 \succ T_3$ for ties, and give agent 1 its complement.

The construction uses the unrestricted nonnegative report space of the model; no continuity, monotonicity, or truthfulness condition is imposed on the mechanism.

6.1 The induced menu

Fix a report $w = b_2$, and let

$$W = w(M), \quad h = \max_{H \in \mathcal{E}} w(H).$$

Lemma 4 (Menu characterization). *The attainable menu of agent 1 under Ψ is:*

1. if $w = 0$, precisely the cycle edges $\mathcal{E}$;
2. if $w \neq 0$,

$$\mathcal{D}(w) = \left\{ B \subseteq M : w(B) \leq \min \left\{ \frac{W}{2}, h \right\} \right\}.$$

Proof. The zero-report case follows directly from the definition.

Assume $w \neq 0$, and suppose first that a row (C, H) gives agent 1 the bundle B .

If $T_1 = C$ wins, then $B = M \setminus C$, and

$$w(C) \geq w(M \setminus C) = w(B),$$

so $w(B) \leq W/2$. Moreover,

$$w(C) \geq w(M \setminus H) = W - w(H),$$

which implies $w(B) \leq w(H) \leq h$.

If $T_2 = M \setminus C$ wins, the same argument, with C and its complement exchanged, gives $w(B) \leq W/2$ and $w(B) \leq h$.

Finally, suppose $T_3 = M \setminus H$ wins. Then $B = H$, so $w(B) \leq h$. Since

$$w(M \setminus H) \geq \max\{w(C), w(M \setminus C)\} \geq \frac{W}{2},$$

we also have $w(B) \leq W/2$.

This proves the forward inclusion.

Conversely, let $B \subseteq M$ satisfy

$$w(B) \leq \min\{W/2, h\}.$$

Choose

$$C = M \setminus B$$

and let H^* be a maximum- w cycle edge. Then

$$w(C) = W - w(B) \geq w(B) = w(M \setminus C)$$

and

$$w(C) = W - w(B) \geq W - h = w(M \setminus H^*).$$

Thus the first-priority candidate C wins, and agent 1 receives B . Since every row is selectable, B is attainable. $\square$

We will use the following elementary property of the five-cycle.

Lemma 5 (Cycle domination). *Let v be nonnegative and additive on the vertices of C_5 . If $B \subseteq M$ satisfies*

$$v(B) \geq v(H) \quad \text{for every } H \in \mathcal{E},$$

then B is EFX against $M \setminus B$.

Proof. Let $K = \max_{H \in \mathcal{E}} v(H)$, so $K \leq v(B)$.

If $|B| \geq 3$, then after removing any good from $M \setminus B$, at most one good remains. Every singleton is contained in a cycle edge, so its value is at most $K \leq v(B)$.

Suppose $|B| = 2$. After removing a good from the complement, at most two goods x, y remain. If they are adjacent, their value is at most K . If they are nonadjacent, the structure of C_5 gives distinct $b_x, b_y \in B$ such that x is adjacent to b_x and y is adjacent to b_y . Since

$$v(x) + v(b_x) \leq v(B) = v(b_x) + v(b_y),$$

we have $v(x) \leq v(b_y)$; similarly, $v(y) \leq v(b_x)$. Hence

$$v(x) + v(y) \leq v(B).$$

If $B = \{b\}$, the two neighbors of b have value zero, since each edge containing b has value at most $v(b)$. The two remaining vertices form an edge, of value at most $K \leq v(B)$. Removing any positively valued good from the complement therefore leaves value at most $v(B)$.

Finally, if $B = \emptyset$, every edge, and hence every good, has value zero, so the EFX condition is vacuous. $\square$

6.2 Equilibrium existence and EFX

Theorem 2. *For every pair of nonnegative additive valuations on five goods, the mechanism Ψ has a pure Nash equilibrium. Moreover, every pure Nash equilibrium of Ψ induces an EFX allocation.*

Proof. We first prove existence.

Suppose $v_2 \neq 0$, and let agent 2 report truthfully:

$$b_2 = v_2.$$

For each of the finitely many rows, truthful reporting gives agent 2 a true-value-maximizing member of

$$\{C, M \setminus C, M \setminus H\}.$$

Choose a row maximizing the resulting utility of agent 1, and let agent 1 report any vector encoding that row. Agent 1 is best-responding by construction. Agent 2 is also best-responding: every deviation can only select one of the same three candidates, including the zero-report outcome $M \setminus H$, and truthfulness already gives her a best candidate. Thus the profile is a PNE.

If $v_2 = 0$, let agent 2 report zero and choose a row whose edge H maximizes $v_1(H)$. Agent 1 then maximizes her utility over the zero-report menu $\mathcal{E}$, while agent 2 is indifferent among all outcomes. This is again a PNE.

We now prove that every PNE is EFX. Fix a PNE $b = (b_1, b_2)$, let

$$(C, H) = \sigma(b_1),$$

and denote the resulting allocation by (B, T) , where B is agent 1's bundle.

For agent 2, if $v_2 = 0$, EFX is immediate. Otherwise, truthful reporting selects a true-value-maximizing bundle among

$$C, \quad M \setminus C, \quad M \setminus H.$$

At equilibrium, T must have the same maximum true value; otherwise truthful reporting would be a profitable deviation.

If $T = C$ or $T = M \setminus C$, its complement is itself one of the candidates, so agent 2 is envy-free. If $T = M \setminus H$, then

$$v_2(T) \geq \max\{v_2(C), v_2(M \setminus C)\} \geq \frac{v_2(M)}{2}.$$

Thus $v_2(T) \geq v_2(H)$, and agent 2 is again envy-free. In particular, she is EFX.

It remains to consider agent 1.

If $b_2 = 0$, Lemma 4 says that her attainable bundles are precisely the cycle edges. At equilibrium, B maximizes v_1 over those edges. Therefore $v_1(B)$ dominates every cycle edge, and Lemma 5 implies that B is EFX.

Now suppose $b_2 = w \neq 0$, and put

$$\tau = \min \left\{ \frac{W}{2}, h \right\}.$$

By the PNE condition and Lemma 4, B maximizes v_1 over all bundles of w -weight at most τ .

First suppose $\tau = W/2$. Let $g \in T$ satisfy $v_1(g) > 0$, and set

$$X = T \setminus \{g\}, \quad Y = B \cup \{g\}.$$

The sets X, Y partition M , so

$$w(X) + w(Y) = W,$$

and at least one of them has weight at most $W/2$. If X is feasible, optimality of B gives

$$v_1(X) \leq v_1(B).$$

If Y is feasible, then

$$v_1(Y) = v_1(B) + v_1(g) > v_1(B),$$

contradicting optimality. Hence the former alternative must hold, which is exactly EFX.

Finally, suppose $\tau = h < W/2$. Every cycle edge H' satisfies

$$w(H') \leq h = \tau$$

and is therefore attainable. Optimality gives

$$v_1(B) \geq v_1(H') \quad (H' \in \mathcal{E}).$$

By Lemma 5, B is EFX. Thus both agents are EFX at every PNE. $\square$

If qualifying mechanisms are required to be defined simultaneously for every number of goods, define the hybrid mechanism

$$\widehat{\Psi}_M = \begin{cases} \Psi, & |M| = 5, \\ \text{Mod-Cut\&Choose}, & |M| \neq 5. \end{cases}$$

Amanatidis et al. [2] prove that Mod-Cut&Choose admits a PNE for every two-agent instance and that all of its PNE are MMS and EFX. Together with Theorem 2, this makes $\widehat{\Psi}$ a qualifying mechanism family. If mechanisms are understood on a fixed ground-set domain, the hybrid extension is unnecessary.

7 An equilibrium attaining 5/6

Consider the true valuations

	0	1	2	3	4
v_1	3	2	3	2	2
v_2	1	1	1	1	1

and the row

$$C = H = \{0, 1\}.$$

Let agent 1 submit any report encoding this row, and let agent 2 submit the zero report. The mechanism returns

$$A_1 = \{0, 1\}, \quad A_2 = \{2, 3, 4\}.$$

Against agent 2's zero report, agent 1 can obtain only a cycle edge. Their v_1 -values are

$$5, 5, 5, 4, 5,$$

so her current bundle is optimal.

With the row fixed at $C = H = \{0, 1\}$, every report of agent 2 gives her either

$$\{0, 1\} \quad \text{or} \quad \{2, 3, 4\}.$$

Under her unit valuation, the current size-three bundle is optimal. Thus the specified profile is a PNE.

Agent 1's equilibrium utility is

$$v_1(A_1) = 5.$$

Her total value is

$$v_1(M) = 12,$$

so $\mu_1 \leq 6$. On the other hand, the partition

$$\{0, 2\} \mid \{1, 3, 4\}$$

has values

$$v_1(\{0, 2\}) = 6, \quad v_1(\{1, 3, 4\}) = 6.$$

Therefore $\mu_1 = 6$, and the equilibrium ratio is exactly

$$\frac{v_1(A_1)}{\mu_1} = \frac{5}{6}.$$

We have thus proved the matching upper bound

$$\alpha^* \leq \frac{5}{6}.$$

8 Conclusion

Combining Theorem 1 with the five-good equilibrium yields the exact answer.

Theorem 3 (Sharp universal guarantee). *For two agents with nonnegative additive valuations,*

$$\boxed{\alpha^* = \frac{5}{6}}.$$

Equivalently, every PNE of every qualifying mechanism gives every agent at least 5/6 of her maximin share, and this factor cannot be improved uniformly over the class of qualifying mechanisms.

Thus strategic EFX equilibria exhibit a sharp uniform separation from arbitrary EFX allocations: the generic 2/3-MMS guarantee improves to 5/6, and the five-good mechanism shows that this improvement is best possible.

References

- [1] Georgios Amanatidis, Georgios Birmpas, Federico Fusco, Philip Lazos, Stefano Leonardi, Rebecca Reiffenhäuser. *Allocating Indivisible Goods to Strategic Agents: Pure Nash Equilibria and Fairness*. Mathematics of Operations Research, 2023. <https://doi.org/10.1287/moor.2022.0058>
- [2] G. Amanatidis, G. Birmpas, F. Fusco, P. Lazos, S. Leonardi, and R. Reiffenhäuser. Allocating indivisible goods to strategic agents: Pure Nash equilibria and fairness. *arXiv preprint arXiv:2109.08644*, 2021. <https://arxiv.org/abs/2109.08644>.